Sequences & Series

ANSWER KEY

Section 1: Arithmetic Sequences & nth Term

1 In an arithmetic sequence where $a_1 = 5$ and $d = 3$, what is the 12th term a_{12}?

a. 38 ✓

b. 41

c. 35

d. 44

$a_n = a_1 + (n-1)d = 5 + (12-1)(3) = 5 + 33 = 38$.

2 If the 4th term of an arithmetic sequence is 17 and the 8th term is 33, what is the common difference d?

a. 4 ✓

b. 5

c. 3

d. 6

$a_8 - a_4 = (8-4)d \Rightarrow 33 - 17 = 4d \Rightarrow 16 = 4d \Rightarrow d = 4$.

3 What is the sum of the first 10 terms of the arithmetic series: $2 + 5 + 8 + \ldots$?

a. 155 ✓

b. 150

c. 160

d. 145

$a_1 = 2, d = 3$. $a_{10} = 2 + 9(3) = 29$. $S_{10} = \frac{10}{2}(a_1 + a_{10}) = 5(2 + 29) = 5(31) = 155$.

Section 2: Geometric Sequences & Finite Series

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4 What is the 6th term of the geometric sequence $2, 6, 18, 54, \ldots$?

a. 486 ✓

b. 162

c. 1458

d. 243

$a_1 = 2, r = 3$. $a_6 = a_1 r^5 = 2(3^5) = 2(243) = 486$.

5 In a geometric sequence, $a_1 = 4$ and $r = 2$. What is the sum of the first 5 terms (S_5)?

a. 124 ✓

b. 60

c. 120

d. 128

$S_n = \frac{a_1(1-r^n)}{1-r} = \frac{4(1-2^5)}{1-2} = \frac{4(1-32)}{-1} = \frac{4(-31)}{-1} = 124$.

6 What is the common ratio r of a geometric sequence where $a_2 = 6$ and $a_5 = 162$?

a. 3 ✓

b. 2

c. 4

d. 9

$\frac{a_5}{a_2} = r^3 \Rightarrow \frac{162}{6} = 27 \Rightarrow r^3 = 27 \Rightarrow r = 3$.

Section 3: Infinite Geometric Series & Convergence

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7 What is the sum of the infinite geometric series: $12 + 6 + 3 + 1.5 + \ldots$?

a. 24 ✓

b. 18

c. 20

d. 36

$a_1 = 12, r = \frac{6}{12} = \frac{1}{2}$. Since $|r| < 1, S_\infty = \frac{a_1}{1-r} = \frac{12}{1-0.5} = \frac{12}{0.5} = 24$.

8 For what values of x does the infinite geometric series $\sum_{k=1}^{\infty} (2x)^k$ converge?

a. $|x| < \frac{1}{2}$ ✓

b. $|x| < 1$

c. $|x| < 2$

d. $x > 0$

Convergence requires $|r| < 1 \Rightarrow |2x| < 1 \Rightarrow |x| < \frac{1}{2}$.

9 Express the repeating decimal $0.\overline{6} = 0.666\ldots$ as a simplified fraction.

a. $\frac{2}{3}$ ✓

b. $\frac{3}{5}$

c. $\frac{6}{11}$

d. $\frac{1}{3}$

Infinite series with $a_1 = 0.6$ and $r = 0.1$: $S_\infty = \frac{0.6}{1-0.1} = \frac{0.6}{0.9} = \frac{2}{3}$.

Section 4: Sigma Notation & Binomial Theorem

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10 Evaluate the summation: $\sum_{k=1}^{5}(3k-2)$.

a. 35 ✓

b. 30

c. 40

d. 25

For $k = 1, 2, 3, 4, 5$: terms are $1, 4, 7, 10, 13$. Sum $= 1 + 4 + 7 + 10 + 13 = 35$.

11 In the expansion of $(x + y)^5$, what is the coefficient of x^3y^2?

a. 10 ✓

b. 5

c. 20

d. 15

Coefficient is $\binom{5}{2} = \frac{5 \times 4}{2 \times 1} = 10$.

12 How many terms are there in the binomial expansion of $(2a - b)^7$?

a. 8 ✓

b. 7

c. 9

d. 14

The binomial expansion of $(x + y)^n$ always contains $n + 1$ terms: $7 + 1 = 8$ terms.

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